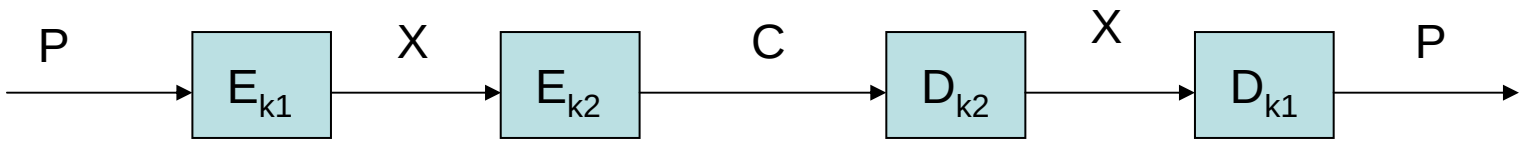
Double DES: Meet in The Middle Attack

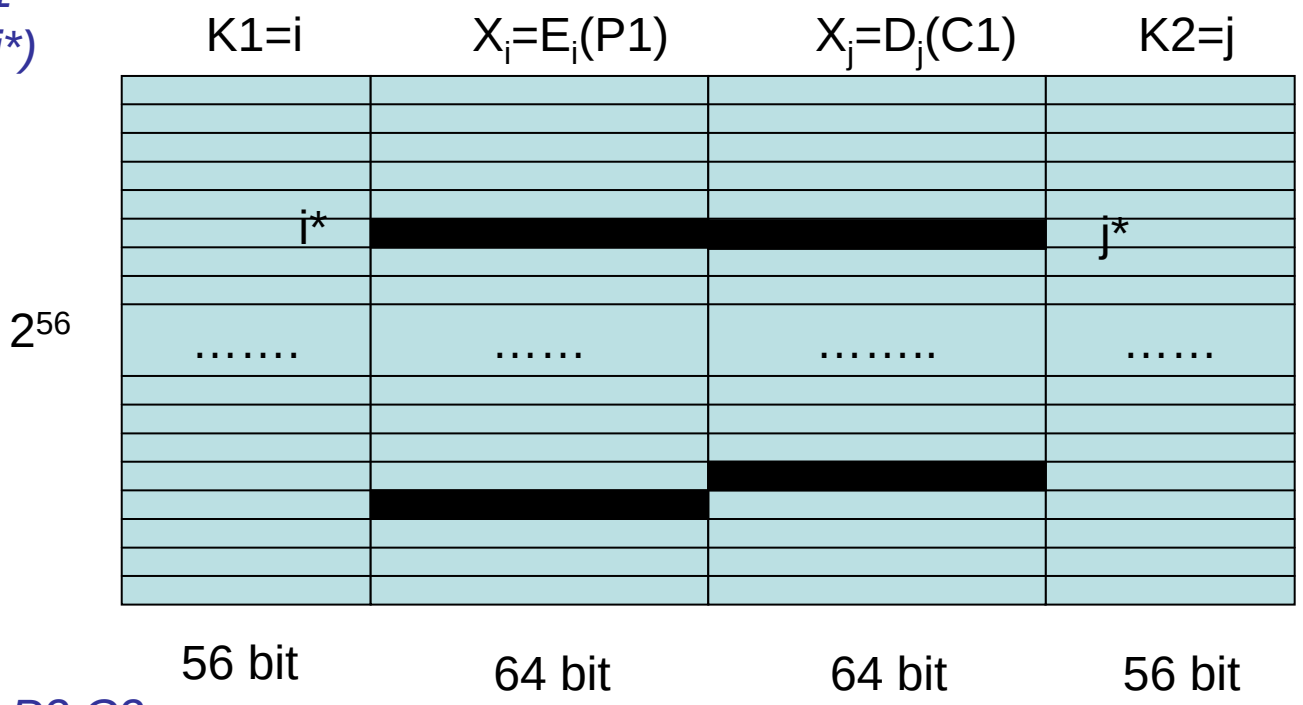
Known Plaintext Attack, two pairs: P1;C1 & P2;C2



$X=E_{k1}(P)$ $X=D_{k2}(C)$

First Pair: P1;C1
To find keys (i;j*)*
- *Xi*= Xj**

Total Memory
$256x2^{56}=2^{64}$bit
2^{61}byte~10^{18}byte
1Tbyte x 10^6run



Second Pair: P2;C2
To check keys (i;j*): P2=Di*[Dj*(C2)]; C2=Ej*[Ei*(P2)]; Average Run # = 2x $2^{56}/2=2^{56}$*